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What is AI Agent Planning?

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AI agent planning refers to the process by which an artificial intelligence (AI) agent determines a sequence of actions to achieve a specific goal. It involves decision-making, goal prioritization and action sequencing, often using various planning algorithms and frameworks.

AI agent planning is a module common to many types of agents that exists alongside other modules such as perception, reasoning, decision-making, action, memory, communication and learning. Planning works in conjunction with these other modules to help ensure that agents achieve outcomes desired by their designers.

Not all agents can plan. Unlike simple reactive agents that respond immediately to inputs, planning agents anticipate future states and generate a structured action plan before execution. This makes AI planning essential for automation tasks that require multistep decision-making, optimization and adaptability.

Advances in large language models (LLMs) such as OpenAI's GPT and related techniques involving machine learning algorithms resulted in the generative AI (gen AI) boom of recent years, and further advancements have led to the emerging field of autonomous agents.

By integrating tools, APIs, hardware interfaces and other external resources, agentic AI systems are increasingly autonomous, capable of real-time decision-making and adept at problem-solving across various use cases.

Complex agents can't act without making a decision, and they can't make good decisions without first making a plan. Agentic planning consists of several key components that work together to encourage optimal decision-making.

Goal definition

The first and most critical step in AI planning is defining a clear objective. The goal serves as the guiding principle for the agent's decision-making process, determining the end state it seeks to achieve. Goals can either be static, remaining unchanged throughout the planning process, or dynamic, adjusting based on environmental conditions or user interactions.

For instance, a self-driving car might have a goal of reaching a specific destination efficiently while adhering to safety regulations. Without a well-defined goal, an agent